How the cluster size is measured

The paper's method (ref. [18]) compared with what you did

ten Wolde, Ruiz-Montero, Frenkel, *J. Chem. Phys.* **104**, 9932 (1996), Sec. III.A and Appendix A
versus `analyze_q6_cluster.py` and the OVITO PTM/CNA analysis

Why this matters. The cluster size n is the only quantity that tells you whether a nucleus formed and how big it is: it is the paper's reaction coordinate, and the number you quote ("critical nucleus ≈ 120") is only meaningful together with the definition behind it.

1 Summary (read this first)

1. The Sanz et al. paper does **not** define its cluster criterion; it only cites ref. [18]. Ref. [18] is a Lennard-Jones paper, and its method is the **q6 bond-order criterion**. Your script follows the same recipe.
2. Your script **differs from ref. [18] in two parameters**, and the difference is in the same direction (yours is stricter):
   - dot-product threshold: **0.70** (yours) vs. **0.5** (paper)
   - cluster-joining rule: yours additionally requires the bond between two solid-like particles to be "connected" (> 0.70); the paper joins any two solid-like *neighbours*.

   The other numbers are the same or close (see the table in section 3).
3. On your own data the result is **sensitive to these thresholds**: one trajectory gave a peak largest cluster of 70, 55 or 40 depending on the setting (section 4). So absolute cluster sizes cannot be compared with the paper's 120 or 65 unless the definition matches.
4. The clean-campaign analysis (PTM/CNA) is a **different kind of quantity**: it counts only fcc/hcp particles, excludes bcc, so it is structure-sensitive. The paper's n is structure-blind and would also count a CsCl nucleus.
5. **Recommendation (corrected after the re-analysis in note 06):** the ref. [18] thresholds (0.5, 1.5) call about 54 of 1000 particles solid-like in this liquid, against the paper's "less than five", so they are *not* usable unchanged. Your stricter settings (0.70, 1.45) give about 1 and fit the paper. Calibrate the thresholds on the liquid first, state them with every cluster size, and keep PTM to *identify the structure*, not to define n (section 6).

2 The paper's method, step by step (ref. [18])

All quotes and numbers below are from the PDF you provided.

Step 1: neighbours and bond orientations

For each particle i, the neighbours are all particles j within a radius r_q. The vectors $\mathbf{r}_{ij}$ are "bonds", with unit vectors $\hat{\mathbf{r}}_{ij}$ and angles (θ_{ij}, ϕ_{ij}). $r_q = 1.5\,\sigma$, chosen "approximately [at] the first minimum of $g(r)$ in a fcc crystal at coexistence" (Lennard-Jones, so this value belongs to that system).

Step 2: local order vector (Eq. A1)

$$\bar{q}_{lm}(i) = \frac{1}{N_b(i)} \sum_{j=1}^{N_b(i)} Y_{lm}(\hat{\mathbf{r}}_{ij}), \qquad l = 6,\ m = -6, \ldots, 6. \tag{1}$$

This is a 13-component complex vector for each particle, the average of the spherical harmonics over its bonds.

Step 3: normalise (Eq. 15)

$$\tilde{q}_{6m}(i) = \frac{\bar{q}_{6m}(i)}{\left[\sum_{m=-6}^{6} |\bar{q}_{6m}(i)|^2\right]^{1/2}}, \qquad \text{so} \quad \mathbf{q}_6(i) \cdot \mathbf{q}_6(i) = 1. \tag{2}$$

Step 4: connect neighbours (Eq. 16)

$$\mathbf{q}_6(i) \cdot \mathbf{q}_6(j) = \sum_{m=-6}^{6} \tilde{q}_{6m}(i)\, \tilde{q}_{6m}(j)^*. \tag{3}$$

Particles i and j are **connected** if this dot product exceeds a threshold, **0.5** in the paper.

Step 5: decide who is solid-like

A single connection is not enough: "even in the liquid it will frequently happen that the bond-order of neighbouring particles is in phase". A particle is therefore solid-like if its **number of connections exceeds a threshold**; they use **seven**. Calibration (Lennard-Jones, Fig. 2 of ref. [18]): with this threshold more than 99% of fcc particles, more than 97% of bcc particles and less than 1% of liquid particles are classed as solid-like. (The text says "threshold value of seven" and "exceeds"; whether the test is ≥ 7 or > 7 is not stated precisely.)

Step 6: build clusters

"Any two solid-like particles that are neighbours belong to the same solid cluster." The cluster size n is the number of particles in the largest cluster.

Properties worth remembering. The method is local (per particle), uses *no* information about particle type, and is *"rather insensitive to the crystal structure of the cluster"* (their words). That is why the Sanz paper can say that disordered fcc and CsCl particles both count as crystalline. The smooth weight $a(r)$ of Eq. A6 is only needed for constraint molecular dynamics; for MC it is not used.

What the Sanz paper adds

Only two statements: at $T^* = 1$, $p^* = 15$ every particle in disordered fcc *or* CsCl is identified as crystalline, and in the metastable liquid "less than five out of 1000 particles" are. **It does not say whether the thresholds of ref. [18] were re-tuned for the binary Yukawa system.** That is the one thing neither document tells you.

3 What you did, side by side

Your q6 analysis is `analyze_q6_cluster.py` (identical copies in `1_step_50K`, `4_step_1.5M_to_3.5M` and `3_Tests_for_codes/1_test_q6_py_code`). The command stored in `cluster_analyze_py` is `python3 analyze_q6_cluster.py traj.lammpstrj --rcut 1.45 --dot 0.70 --min-solid-bonds 7`.

Ingredient	Ref. [18]	Your q6 script	Match?
Bond-order degree l	6	6	same
Local vector	average of Y_{6m} over bonds (A1)	mean of Y_{6m} over neighbours	same
Normalisation	unit vector (Eq. 15)	normalised to unit length	same
Dot product	$\sum \tilde{q}\,\tilde{q}^*$ (Eq. 16)	real part of the vector product	same
Neighbour cutoff r_q	$1.5\,\sigma$ (first minimum of $g(r)$, LJ fcc)	$1.45\,\sigma$	close
Connection threshold	**0.5**	**0.70**	different
Connections for "solid-like"	7	≥ 7	same
Cluster joining	any two neighbouring solid-like particles	solid-like *and* bond dot product > 0.70	different
Particle types	ignored (single species)	ignored (type not used)	same
Thresholds tuned on	LJ fcc/bcc/liquid at coexistence	not recorded	not stated

Direction of the effect. A higher connection threshold (0.70 vs. 0.5) means fewer bonds count as connected, so fewer particles reach 7 connections, and the extra joining condition removes some links between solid-like particles. Both make your clusters *smaller or equal* to what the paper's criterion would give on the same configuration. (This is expected from the definitions; I have *not* measured it on your data.)

Is the cutoff sensible for your system? Simple geometry (my estimate, not a simulation result): at the density of your liquid parent, $\rho\sigma^3 \approx 1.037$, an fcc lattice has a nearest-neighbour distance of about $1.11\,\sigma$ and a second shell at about $1.57\,\sigma$, so a first-shell cutoff between them such as 1.45 is reasonable. A bcc lattice at the same density has shells at about $1.08\,\sigma$ and $1.25\,\sigma$, so with 1.45 both shells fall inside the cutoff (about 14 neighbours) – different from fcc's 12. The paper's value of 1.5 was fixed for fcc LJ; for a CsCl crystal the right cutoff is not obvious.

The PTM/CNA analysis in the clean campaign

`analyze_campaign.py` and `ovito_structure.py` do something different:

1. OVITO classifies each particle (PTM with RMSD cutoff 0.10, and adaptive CNA) as fcc, hcp, bcc, icosahedral, sc or other.
2. The reported "largest FCC/HCP cluster" joins **fcc and hcp particles only** that are within $1.45\,\sigma$ of each other.

So this count (i) **excludes bcc** and therefore would not count a CsCl nucleus, and (ii) depends on how well PTM recognises thermally distorted small clusters. It is very good at *naming a structure* and weaker as a structure-blind *size* measure. Your own notes already say PTM/CNA "can miss very small or strongly distorted embryos" (`KNOWN_LIMITATIONS.md`, item 4).

4 Evidence from your own data

4.1 Does your criterion call the liquid liquid? (paper: < 5 of 1000)

Particles classed as solid-like per saved frame, default settings (from the `cluster_size.dat` files):

Run	Frames	Mean solid-like	Max solid-like
`1_step_50K`	51	1.4	5
`2_step_50K_to_500K`	451	1.3	6
`3_step_500K_to_15000K`	1001	2.1	13

In the early liquid this matches the paper's statement (fewer than 5 per 1000, on average about 1). Later in run 3 a few frames exceed that because a small cluster is starting to form.

4.2 How much do the thresholds change the answer?

Same trajectory (`4_step_1.5M_to_3.5M`: $N = 1000$, $T^* = 1$, $p^* = 18$, dr = 0.01, vmax = 0.005, seed 68593, 2×10^6 cycles from the 1.5M configuration), three analyses:

Setting	rcut / dot / min. bonds	Peak largest cluster	Peak at cycle
Default	1.45 / 0.70 / 7	**70**	1.946M
Stricter bonds	1.45 / 0.75 / 7	40	1.976M
More connections	1.45 / 0.70 / 8	55	1.976M

Largest cluster along the default analysis at 0, 0.25, 0.5, …, 2.0M cycles: 1, 1, 5, 3, 6, 8, 16, 21, 41.

Take-away. On one and the same trajectory the peak cluster size moves between 40 and 70 depending on the threshold, i.e. by a factor of about 1.75. Numbers like "120" or "65" from the paper can only be compared to yours if the definition is the same. *Trends* (growth, plateau) are robust; *absolute sizes* are not.

4.3 The unexplained difference between your two eras

	`2_1000_atoms` (June)	`4_Clean`, dr = 0.01	`4_Clean`, dr = 0.04
dr	0.01	0.01	0.04
vmax	**0.005**	0.0005	0.0005
Start	ramp-prepared liquid, density label 1.044	validated parent ($\rho = 1.037$)	same parent
Cluster measure	q6 (0.70 / 7)	PTM fcc+hcp	PTM fcc+hcp
Largest cluster	≈ 70	≤ 13	≈ 500

The two dr = 0.01 results disagree (70 vs. 13), but I cannot say why. Four things changed at once: the cluster measure, vmax (10×), the starting configuration and the random seed. The data on disk do not separate them. What the notes do record: `PHASE_02_LIQUID_VALIDATION.md` says the earlier q6 analysis "was not trusted because an apparent cluster was not obvious during visual inspection", which is why the clean campaign switched to PTM/CNA. You had marked those frames (`marked_1946000.lammpstrj`, `marked_1976000_dot075.lammpstrj`), but I found no structural analysis (for example PTM) of them. So the 70-particle cluster is best read as *not confirmed as a crystal*: it may have been a locally ordered liquid fluctuation that the q6 criterion counted.

5 What is unknown

- Whether Sanz et al. re-tuned the ref. [18] thresholds (0.5, 7, $r_q = 1.5$) for the binary Yukawa system. Not stated in either paper.
- Whether "7 connections" means ≥ 7 or > 7 in ref. [18].
- What the best cutoff is for CsCl/bcc crystals in your system.
- Why the June q6 run (70) and the clean PTM run (13) differ at dr = 0.01.
- Whether the June 70-particle cluster was a real crystal: it was distrusted after visual inspection, but no PTM/CNA of those frames is recorded.

6 Recommendation

1. **Use the ref. [18] *method* but calibrate the thresholds on the liquid.** Note 06 shows that its LJ values (0.5, 1.5) over-count in this system (about 54 false solid-like particles per 1000); require fewer than 5 in the validated liquid, as the paper does. Your 0.70 / 1.45 / 7 satisfies this. State all the numbers whenever you quote a cluster size.
2. **Keep a second setting (for example 0.75 or 8 bonds) as a sensitivity check**, so that you can say how much n depends on the definition. That is the 40–70 spread in section 4.2.
3. **Use PTM/CNA to name the structure** (fcc/hcp/bcc fractions in the nucleus), not to define n. Add a charge-resolved analysis to test "disordered" versus "ordered".
4. **Say in the meeting** that absolute cluster sizes are definition-dependent and that your replication target at $p^* = 18$ is qualitative (the paper gives no cluster sizes there).

One optional step that needs your OK. To close the gaps in section 5 I would re-analyse *existing* trajectories (no simulation): run your script with the ref. [18] settings on the June trajectory and on the dr = 0.04 clean-campaign trajectories, and put the q6 size next to the PTM size frame by frame. This is post-processing only, but it is a computation on your data, so I have not done it.